

THE LITTLE LION

Leo Minor



SIZE RANKING 64

BRIGHTEST STAR
46 Leonis Minoris 3.8GENITIVE
Leonis Minoris

ABBREVIATION LMi

HIGHEST IN SKY AT 10 P.M.
March–AprilFULLY VISIBLE
90°N–48°S

This small, insignificant constellation, adjacent to Leo in the northern sky, represents a lion cub, although this is not suggested by the pattern of its stars. It was introduced in the 17th century by the Polish astronomer Johannes Hevelius (see p.384).

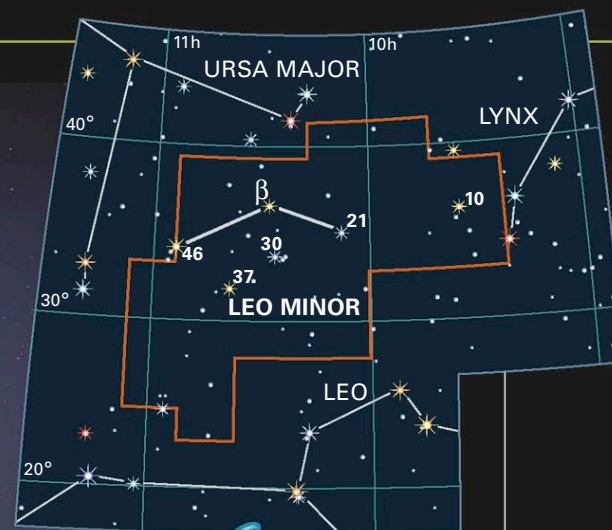
SPECIFIC FEATURES

Unusually, this constellation has no star labeled Alpha. This is due to an error by the 19th-century English astronomer Francis Baily, who assigned the Greek letters to the constellation's stars. When doing so, he overlooked assigning a Bayer letter to the brightest star, 46 Leonis Minoris, which should have been recorded as Alpha (α), although he did label the second-brightest star as Beta (β) Leonis Minoris.

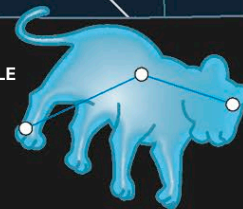
Although Leo Minor contains no objects of interest for users of binoculars or a small telescope, Beta (β) is a close double star that can be separated by a telescope with very large aperture. It has a magnitude of 4.2, and its component stars orbit each other every 37 years.

THE LION CUB

Having located the distinctive shape of the Sickle in Leo (top, right), look north of it to find the faint stars of Leo Minor.



THE LITTLE LION



BERENICE'S HAIR

Coma Berenices



SIZE RANKING 42

BRIGHTEST STAR
Beta (β) 4.2GENITIVE
Comae Berenices

ABBREVIATION Com

HIGHEST IN SKY AT 10 P.M.
April–MayFULLY VISIBLE
90°N–56°S

Coma Berenices represents the flowing locks of Queen Berenice of Egypt, which she cut off as a tribute to the gods after the safe return of her husband, Ptolemy III, from battle during the 3rd century BCE. It is a faint but interesting northern constellation, lying between Leo and Boötes. In the mid-16th century, it was named as a constellation by the German cartographer Caspar Vopel. Before then, its stars were regarded as forming the tail of Leo.

SPECIFIC FEATURES

The Coma Star Cluster, also known as Melotte 111, is the constellation's main feature. It comprises several dozen faint stars, which fan out distinctively for several diameters of a full moon southward from Gamma (γ) Comae Berenices. This open cluster, which is seen to best advantage through binoculars, has been imagined as both the bushy tip of a lion's tail and a lock of Berenice's hair.

Coma Berenices contains numerous galaxies in its southern half. Most of these are members of the Virgo Cluster, such as M85, M88, M99, and M100, but two notable exceptions, M64 (see p.314) and NGC 4565, are closer to the Earth.

Popularly known as the Black Eye Galaxy, M64 is a spiral galaxy tilted at an angle to the Earth, which is seen as

an elliptical patch of light through a small telescope; it is best seen with a telescope with an aperture of 6 in (150 mm) or more.

A dust cloud near the galaxy's nucleus creates the "black eye" effect.

NGC 4565, another spiral galaxy, lies edge-on to the Earth and is more difficult to spot. It appears long and thin when viewed through a telescope with a 4-in (100-mm) aperture, and a lane of dark dust is revealed in long-exposure photographs.



NGC 4565

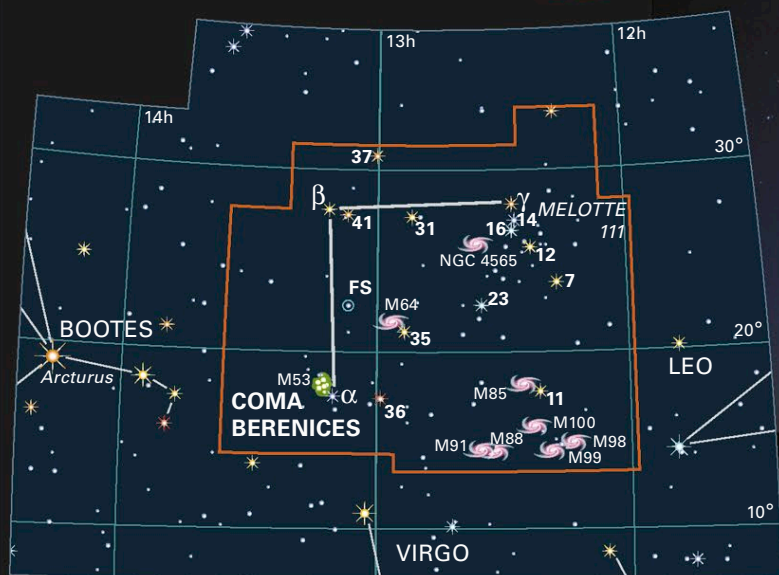
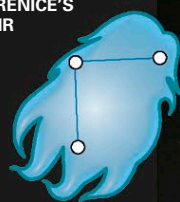
Seen edge-on, this spiral galaxy displays a lane of dark dust along its spiral arms when viewed through larger apertures.

THE BLACK EYE GALAXY

The spiral galaxy M64 sports a large, dark dust cloud near its core, giving it the appearance of a blackened eye.



BERENICE'S HAIR



MANE OF HAIR

The distinctive splay of the Coma Star Cluster marks out Coma Berenices in the night sky. Leo's hindquarters can be seen closer to the horizon.